

Adaptive Signal Acquisition

Living Technical Report - Stages 1 through 3

Purpose

This report is the project record. It explains the mathematical model, implementation choices, controlled experiments, measured results, limitations, and next decision. It must be updated whenever a new stage changes the evidence.

Current evidence-based conclusion

The first information-guided policy is implemented, reproducible, and fair against fixed and random sensing. Across 100 trials in the current direct-sparse setting, it does not beat the strongest non-adaptive baseline at any tested budget. This is a valid finding, not a failure to hide. The next work is a controlled Stage 4 study and ablation, not an unprincipled attempt to tune until adaptive wins.

Stage	Status	Main evidence
1	Complete	Deterministic simulator and visual sanity checks
2	Complete	LASSO reconstruction, fixed/random paired baselines
3	Complete	Sequential information policy and 100-trial comparison
4	Next	Controlled sweeps, ablations, confidence intervals

1. Research question and model

1.1 Project question

When only a small number of noisy measurements can be taken, can selecting each new measurement using earlier observations reconstruct a sparse signal more accurately than selecting all measurements in advance?

The operational goal is lower normalized mean squared error (NMSE) under the same measurement budget, hidden signal distribution, candidate measurements, noise draw, and final reconstruction algorithm.

1.2 Sparse hidden signal

The current simulator uses a directly sparse vector x in $\mathbb{R}^{64}$. It has exactly $k=4$ non-zero coordinates. The active locations are sampled uniformly without replacement and their amplitudes are Gaussian with standard deviation 1.0. The estimator does not receive x or the support locations.

More generally, a signal may be sparse only after a known transform Ψ , written $x = \Psi z$ with $\|z\|_0 = k$. The present implementation uses $\Psi = I$, so z and x are the same vector.

$$x \in \mathbb{R}^N, \quad \|x\|_0 = k \ll N, \quad N = 64, \quad k = 4$$

1.3 Noisy linear measurements

A measurement is not a direct sample of one coordinate. It is a weighted linear combination of all 64 coordinates. A finite dictionary D contains 128 allowed normalized measurement rows. Selecting B rows from D forms the sensing matrix A in $\mathbb{R}^{(B \times 64)}$.

For selected action a_t , the environment returns one noisy scalar y_t . Stacking all selected actions and observations gives $y = A x + \epsilon$.

$$y_t = a_t^T x + \epsilon_t, \quad \epsilon_t \sim N(0, \sigma^2), \quad \sigma = 0.10$$
$$A_B = [a_1^T; a_2^T; \dots; a_B^T], \quad y_{1:B} = A_B x + \epsilon$$

2. Stage 1 - deterministic foundations

2.1 What was built

Stage 1 made every object in the model concrete. A seeded random-number generator creates the sparse signal, the 128-by-64 candidate dictionary, a fixed list of 16 selected actions, and a Gaussian noise vector. Repeating the seed reproduces the same arrays exactly.

The initial example used x with shape (64,), the candidate dictionary with shape (128, 64), A with shape (16, 64), and y with shape (16,). At this stage there was no reconstruction and no adaptive selection.

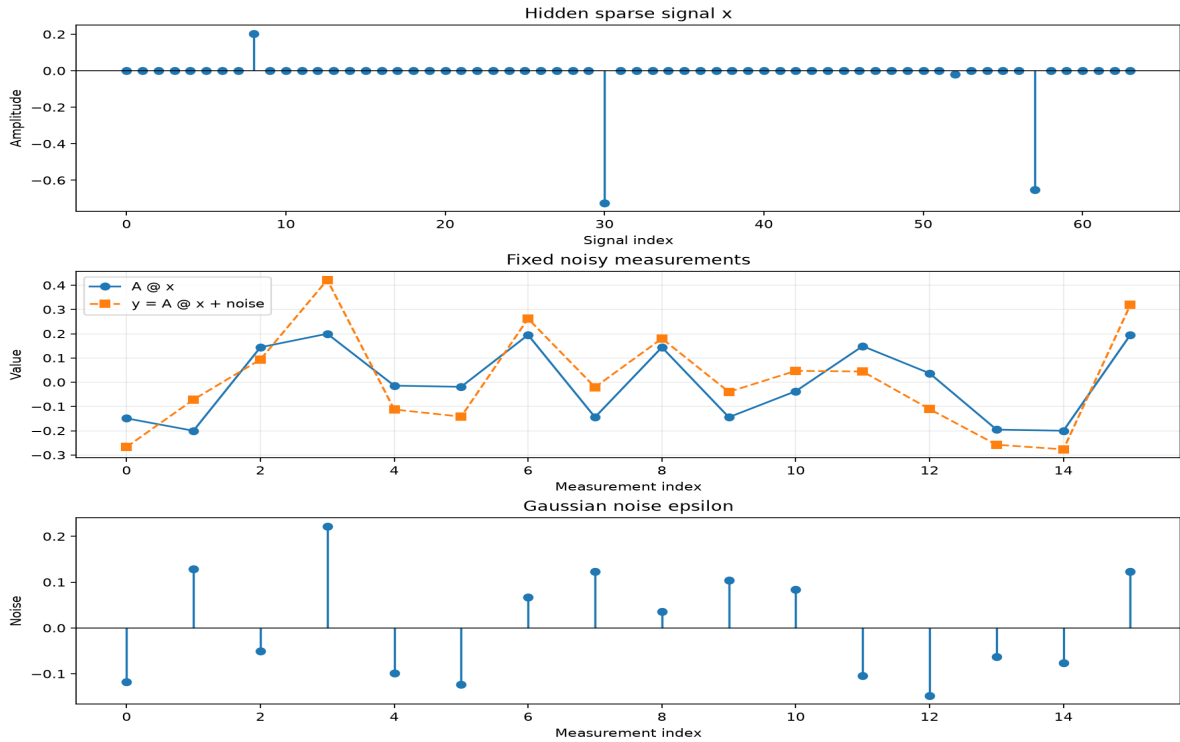


Figure 1. Stage 1 visual sanity check: sparse signal, clean measurements, noise, and observations.

2.2 Why deterministic simulation matters

A stochastic experiment is not trustworthy if a result cannot be reproduced. Fixed seeds let us distinguish a code change from normal randomness. They also ensure a later adaptive method can be compared against the same hidden signal and the same noise realization.

3. Stage 2 - reconstruction and non-adaptive reference

3.1 Final reconstruction with LASSO

After B measurements, LASSO reconstructs a candidate signal z . The first term rewards agreement with observations; the second term encourages most coordinates of z to become exactly zero. In this project z is an estimate of x because the signal is directly sparse.

The solver is ISTA. Each iteration moves z toward lower data mismatch and then applies soft-thresholding, which shrinks weak coefficients to zero. The final LASSO configuration uses $\lambda=0.08$, maximum 5000 iterations, and tolerance $1e-7$.

$$\hat{x} = \arg \min_z [0.5 ||A z - y||_2^2 + \lambda ||z||_1]$$

3.2 Fixed and random baselines

Fixed sensing uses one predetermined permutation of the 128 candidate rows and takes its first B rows in every trial. Random sensing draws a new random row order before a trial begins, then takes its first B rows. Neither method looks at y before deciding its rows.

For each trial and budget, fixed and random receive the identical hidden x and the identical pre-drawn noise prefix. They differ only in selected A rows. This paired design is essential for a fair comparison.

3.3 Evaluation metrics

NMSE is the main reconstruction metric. Lower is better. The $1e-12$ guard prevents division by zero in an all-zero reference. Support F1 measures whether the method identifies non-zero coordinates, using an estimate magnitude threshold of $1e-3$.

$$\text{NMSE} = ||x - \hat{x}||_2^2 / (||x||_2^2 + 1e-12)$$

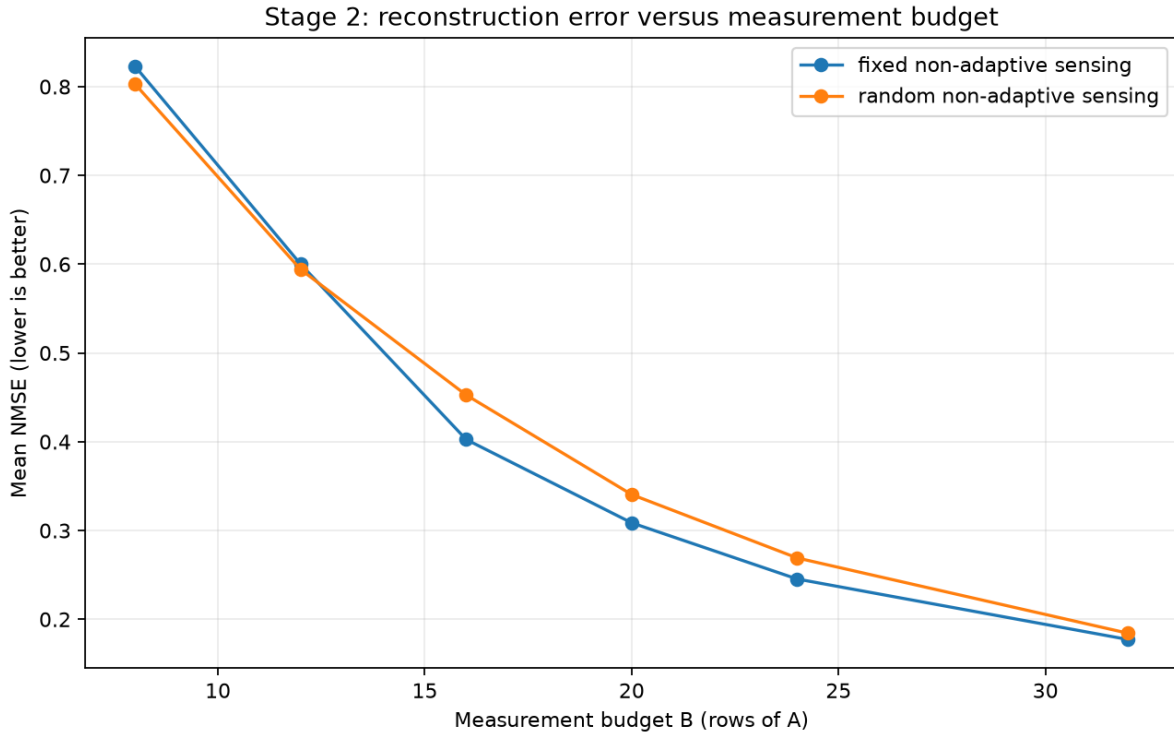


Figure 2. Stage 2 100-trial baseline result: error decreases plausibly as budget increases.

4. Stage 3 - sequential information-guided sensing

4.1 Sequential policy

Unlike the two Stage 2 methods, Stage 3 chooses one action at a time. Before selecting action $t+1$, it has history $h_t = (A_t, y_{1:t})$: all earlier actions and observed values. It does not have the true hidden signal x .

The selected action is appended to A ; earlier rows are never changed. The simulator produces its observation only after the policy has selected that row.

$$a_{(t+1)} = \text{pi}(h_t), \quad h_t = (A_t, y_{1:t})$$

4.2 Gaussian uncertainty approximation

The policy begins with mean $\mu_0 = 0$ and covariance $\Sigma_0 = 0.0625 \mathbf{I}$. The prior variance 0.0625 is a simple heuristic based on the current signal generator: expected active fraction $k/N = 4/64$ with active amplitude variance near 1.

For a possible unused row a , $a^T \Sigma a$ estimates uncertainty in the measurement direction. Relative to noise variance σ^2 , the policy scores it using the linear-Gaussian mutual-information expression below and takes the largest score.

$$\text{score}(a) = \log(1 + (a^T \Sigma a) / \sigma^2)$$

4.3 Update after observing y_t

Once a row is measured, a standard Gaussian conditioning update changes the mean and covariance. K_t is the gain: it is larger in directions that were uncertain and smaller when measurement noise is large. The covariance update reduces uncertainty in the direction that was just measured.

$$K_t = \Sigma_t a_t / (a_t^T \Sigma_t a_t + \sigma^2)$$

$$\mu_{(t+1)} = \mu_t + K_t (y_t - a_t^T \mu_t)$$

$$\Sigma_{(t+1)} = \Sigma_t - K_t (a_t^T \Sigma_t)$$

5. Why the Stage 3 policy uses a support-aware proxy

5.1 The important limitation

In a pure linear-Gaussian model, the covariance update depends on which rows were selected, but not directly on the numerical values observed. That is not sufficiently responsive to sparse evidence. A large or small y value should affect what we consider worth measuring next.

Therefore the current implementation computes a provisional LASSO estimate $\mathbf{z}_{\text{hat}_t}$ after each observation using only $\mathbf{A}_t$ and $\mathbf{y}_{1:t}$. This is not the final reconstruction and it is not an exact sparse Bayesian posterior. It is a transparent heuristic for making the score data-dependent.

$$\mathbf{z}_{\text{hat}_t} = \arg \min_{\mathbf{z}} [0.5 \|\mathbf{A}_t \mathbf{z} - \mathbf{y}_{1:t}\|_2^2 + \lambda \|\mathbf{z}\|_1]$$

5.2 Support-aware score

The policy adds a positive semidefinite term $\gamma \mathbf{z}_{\text{hat}_t} \mathbf{z}_{\text{hat}_t}^T$ to the Gaussian covariance. The default γ is 1. Rows aligned with the provisional sparse estimate receive an increased score. Importantly, this estimate arises only from already observed measurements; the action-selection function has no `hidden_signal` or `x` argument.

$$\begin{aligned} \Sigma_{\text{tilde}} &= \Sigma_t + \gamma \mathbf{z}_{\text{hat}_t} \mathbf{z}_{\text{hat}_t}^T \\ \mathbf{a}_{(t+1)} &= \arg \max_{\text{over unused } \mathbf{a}} \log(1 + (\mathbf{a}^T \Sigma_{\text{tilde}} \mathbf{a}) / \sigma^2) \end{aligned}$$

5.3 What is recorded

For every adaptive step, the code writes budget, trial, step, selected candidate index, observed value, and selected information score. These action traces make the policy inspectable rather than a black box.

6. Stage 3 controlled result

6.1 Protocol

The primary Stage 3 run used 100 independent trials, $N=64$, $k=4$, 128 candidate rows, noise standard deviation 0.1, budgets $B = 8, 12, 16, 20, 24, 32$, and the same final LASSO for every method. Fixed, random, and information-guided sensing used matched hidden signals and matched noise prefixes within each trial and budget.

The table reports mean NMSE. Lower is better. Values are from `artifacts/stage3_adaptive/adaptive_summary.csv` with seed 20260913.

B	Fixed	Random	Information-guided	Best baseline
8	0.8324	0.8238	0.9006	Random
12	0.5616	0.6010	0.7795	Fixed
16	0.4045	0.4185	0.6499	Fixed
20	0.3024	0.3396	0.5078	Fixed
24	0.2400	0.2792	0.4028	Fixed
32	0.1815	0.2004	0.2724	Fixed

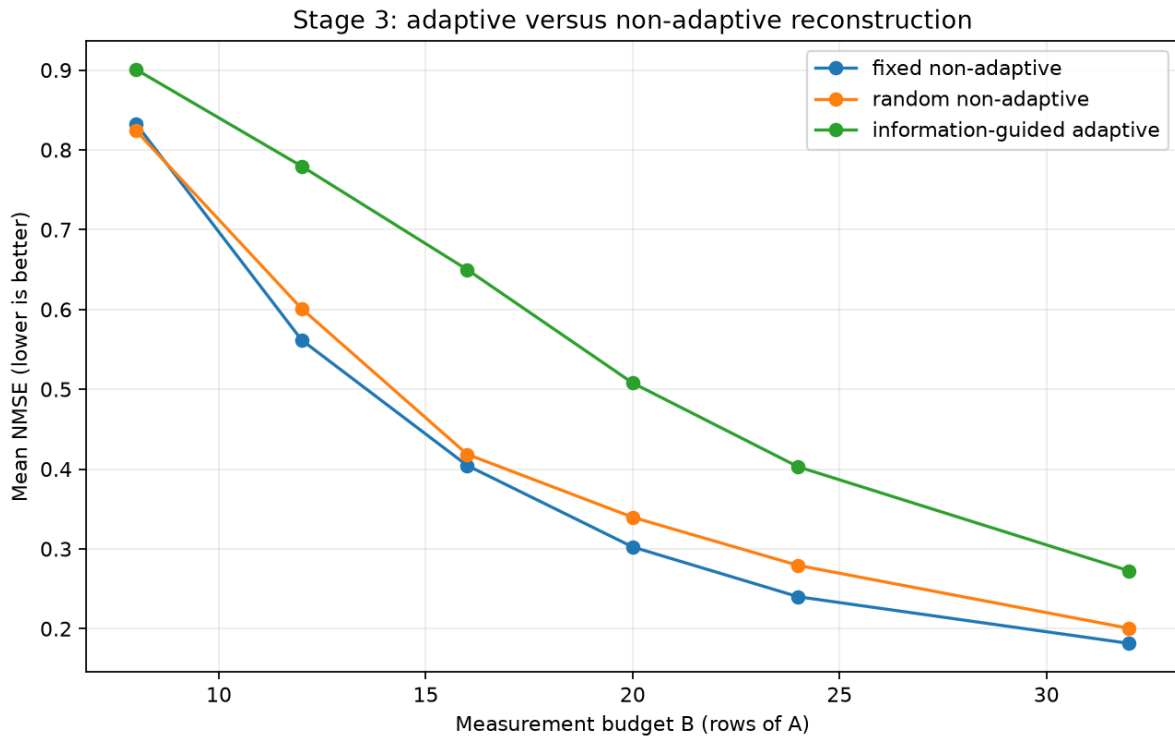


Figure 3. Stage 3 100-trial result: the first information-guided policy has higher mean NMSE than the best baseline at every tested budget.

6.2 Interpretation

The result answers the current question for this exact policy and setting: the first information-guided proxy does not improve final sparse reconstruction. At $B=32$, its mean NMSE is 0.2724, compared with 0.1815 for fixed sensing. This is a meaningful gap, not just a visual tie.

The result does not prove that all adaptive sensing is worse. It says that this particular greedy Gaussian-plus-provisional-LASSO policy is not better in this direct-sparse, dense-random-dictionary regime.

7. Diagnosis and Stage 4 decision

7.1 Why information-guided selection is not guaranteed to win

The information score optimizes approximate uncertainty reduction under a Gaussian model. The project evaluates final NMSE after sparse LASSO reconstruction. Those objectives are related but not identical.

The true signals are hard-sparse, whereas the Gaussian belief is dense. The early provisional LASSO estimate is based on very few measurements and can be unstable; its support-aware term can amplify an early wrong guess. The policy is greedy, optimizing the next measurement score rather than the final B -measurement reconstruction. Finally, all candidate rows are dense random mixtures, so an apparently informative row may not improve the conditioning or support-identification ability of the eventual sensing matrix.

7.2 Correct next move

Proceed to Stage 4, but preserve the present method as `adaptive_v1`. Do not silently modify it until it wins. Stage 4 exists specifically to learn when adaptivity helps, including the possibility that it does not help here.

First run ablations: $\gamma=0$ (Gaussian action-history-only policy), γ values such as 0.1, 0.5, and 1.0, and the present $\gamma=1.0$ policy. This separates the effect of the support-aware proxy from the base uncertainty score.

Then sweep budget, noise/SNR, sparsity k , and signal family. Add one distribution shift, for example calibrating at $k=4$ and testing at $k=8$ or using a different noise distribution. Use 100 trials per condition initially and 500 only for final selected figures. Report bootstrap confidence intervals for paired NMSE differences, not only means.

7.3 What may become `adaptive_v2`

Only after the ablation results identify a problem should we introduce a second policy. Candidate improvements include a sparse posterior or support-probability uncertainty model, a score that explicitly rewards measurement diversity and lower mutual coherence, or a policy selected for expected improvement in sparse reconstruction rather than Gaussian entropy alone. `adaptive_v2` must be a new named method compared against frozen `adaptive_v1` and the original baselines.

Decision: Stage 3 is complete. Stage 4 is the right next stage, and it should begin with controlled ablations rather than an unexplained rewrite of the policy.

8. Reproducibility and maintenance

8.1 Repository map

src/adaptive_signal_acquisition/signals.py defines sparse-signal generation. measurements.py defines the candidate dictionary and $y = A x + \text{noise}$. reconstruction.py contains ISTA LASSO. baselines.py contains fixed/random action selection. sequential.py contains the Stage 3 Gaussian belief, information score, history proxy, and sequential loop. experiments.py creates fair paired studies.

scripts/run_baseline_study.py creates Stage 2 artifacts. scripts/run_adaptive_study.py creates Stage 3 artifacts. scripts/build_living_report.py regenerates this PDF.

8.2 Update rule for this report

After every completed experiment family, update: (1) the exact configuration and seed, (2) equations or policy changes, (3) result tables and figures, (4) the interpretation including failures, and (5) the next decision. Never replace an old result without preserving the method name and configuration that produced it.

The report should remain evidence-led. A conclusion must identify the relevant artifact folder, sample count, metric, and comparison protocol. This is especially important before adding an LLM research orchestrator in later stages.

8.3 Commands

Run tests: `.venv\Scripts\python.exe -m pytest`

Run Stage 2: `.venv\Scripts\python.exe scripts\run_baseline_study.py`

Run Stage 3: `.venv\Scripts\python.exe scripts\run_adaptive_study.py`

Regenerate this report: use the bundled runtime or an environment with reportlab installed to run `scripts\build_living_report.py`

Report version: v1 - Stages 1 through 3 complete